Analyzing the Impact of E-sports Tournaments on the Steam Community Market: A Case Study on CS2 Skins and Stickers

Serkan Dağ – 34255

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ABSTRACT

This extensive research project investigates the systematic macroeconomic cross-dependencies governing the Counter-Strike 2 (CS2) digital asset ecosystem, specifically evaluating the quantified impact of professional live esports viewership on item pricing indices. Utilizing an integrated data science pipeline spanning ten years of historical records, data was extracted across three modern streams: a custom programmatic wrapper targeting the Steam Community Market API, a localized BeautifulSoup parsing engine digesting offline Liquipedia tournament calendars to bypass active network firewalls, and a Twitch category scraping architecture.

The analytical workflow advances through four progressive validation layers. First, classical parametric and non-parametric hypothesis tests (Pearson Correlation, Independent Two-Sample T-Test, and One-Way ANOVA) mathematically establish that live esports tournament blocks and social attention spikes act as severe structural volatility catalysts inside the marketplace, with Valve-sponsored Majors generating an extreme pricing variance shock (F-Statistic = 30.3330, P-Value = 1.4144e-13). Second, an unsupervised K-Means model (k = 3, Silhouette Score = 0.464) segmentally partitions the asset matrix into autonomous macroeconomic market states (Stagnant, Transition, and Hype Markets). Third, benchmarking supervised regression models (Random Forest, Gradient Boosting) reveals a predictive baseline ceiling (R-Squared approx 0), serving as empirical validation of the financial Random Walk Hypothesis while node-split evaluations confirm that daily Twitch audience metrics (avg_viewer) retain an absolute dominant feature prominence score exceeding 80%. Fourth, an advanced sequential time-series ARIMAX(1,1,1) model tracks the natural autoregressive momentum of the market index while incorporating tournament timelines as external exogenous regressors.

The resulting out-of-sample flat forecast field and negative R-Squared parameters yield a rigorous validation of the Efficient Market Hypothesis (EMH) within virtual economies: public information shocks are instantaneously absorbed into digital asset derivatives boards, preventing systematic long-term directional arbitrage. To present these macro insights, an interactive Single Page Application (SPA) visual dashboard powered by Chart.js was successfully deployed.

1. MOTIVATION & RESEARCH QUESTIONS

The rapid transformation of modern gaming infrastructures into decentralized financial ecosystems has introduced a new paradigm of alternative commodity trading. Inside games like Counter-Strike 2 (CS2), weapon cosmetic coatings ("skins") and tournament containers function as highly liquid, tradeable assets. These digital items possess independent float values, rarity tiers, and varying supply-demand curves, operating within an internal economy that drives billions of dollars in annual transaction velocity globally. The pricing dynamics of these items resemble traditional high-frequency asset derivatives boards, showing measurable patterns of volatility, resistance lines, and macro trends.

The underlying motivation of this study is driven by two main factors:

- **Academic Curiosity:** Bridging the gap between behavioral economics, data science pipelines, and digital alternative assets. While traditional financial research heavily explores the impact of public sentiment and news flows on stock markets, the correlation between social video streaming density (live human attention) and virtual asset derivatives remains under-researched.
- **Ecosystem Relevance:** The digital skin economy is highly sensitive to external shocks. As an active software engineering researcher and computer science student closely following competitive esports structures, I set out to investigate whether live human attention—measured via Twitch viewership density—and the scheduling of tier-1 professional tournaments inject systematic, measurable changes into asset valuations, or if the market operates as a pure random walk.

Core Research Questions:

1. Is there a statistically verifiable correlation between day-to-day live streaming crowd spikes on Twitch and the valuation shifts of liquid commodities inside the Steam ecosystem?
2. Do active professional tournament calendar dates introduce meaningful structural differences in asset price variances compared to standard off-season regular days?
3. Can separate tiers of esports tournaments (such as Valve-sponsored Majors versus independent S-Tier events) be classified as distinct economic shock waves inside the market?
4. Can unsupervised clustering algorithms identify distinct macro states within a virtual game-asset marketplace?
5. Is a digital commodity market predictable using standard supervised regression frameworks based on external social interest parameters, or does it conform to the Random Walk and Efficient Market hypotheses?

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2. DATA INVENTORIES & COLLECTION PIPELINES

To establish a rigid analytical foundation, three independent data collection pipelines were engineered from scratch to maximize information harvesting while protecting the codebase from network stability issues.

```
==========================================================================
                      DATA ARCHITECTURE BLUEPRINT
==========================================================================
[Pipeline 1: Steam API] ------> dsa210_mega_data.csv -----\
                                                           \
[Pipeline 2: Liquipedia] ----> tournaments.csv ----------> [ merger.py ]
                                                           / (Inner Join)
[Pipeline 3: Twitch API] ----> cs2_gunluk_izleyici.csv ---/
                                              |
                                              v
                              enriched_market_data.csv (Master Matrix)
==========================================================================
```

2.1. Steam Community Market API Extraction Pipeline

The target asset tracking catalog was specified inside itemler.txt. This ledger contained a carefully curated set of highly liquid, representative baseline skins and high-tier collector containers (e.g., *AK-47 | Redline (Field-Tested)*, *AWP | Asiimov*, and rare tournament sticker capsules). A production-ready Python script (import_items.py) utilized custom URL requests targeting the official Steam Community Market endpoints. The engine executed sequential loops over a multi-year chronological path, handling JSON payload extractions, isolating daily mean price values, and outputting the records into dsa210_mega_data.csv.

2.2. BeautifulSoup Liquipedia Scraper Pipeline

Gathering tournament calendar blocks programmatically from live esports wikis often triggers aggressive automated security firewalls, resulting in IP bans. To ensure total pipeline autonomy and execution stability, the targeted historical S-Tier and Major championship schedule boards were archived locally into an offline format (liquipedia.html).

The localized python script (scrape_tournaments.py) used the BeautifulSoup library to parse the raw HTML tree structure. The algorithm scanned table rows, isolated structural event containers, parsed start and completion timestamps, and filtered tournament titles based on their classification level (Major, S-Tier). The resulting structural calendar was cleanly written to tournaments.csv.

2.3. Historical Twitch Viewership Aggregator Pipeline

To extract the continuous stream of social interaction variables, a script (scrape_twitch_viewership.py) retrieved historical database metrics recording the daily average viewer counts (avg_viewer) specifically allocated to the Counter-Strike directory. This process created cs2_gunluk_izleyici.csv, establishing a clean daily time-series log of live human attention variables.

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3. DATA CLEANING & PREPROCESSING ARCHITECTURE

The raw extraction pipelines produced distinct matrices with severe temporal inconsistencies, missing row variables, and structural layout gaps. A centralized data engineering module (merger.py) was deployed to build the unified master analytical framework.

3.1. Construction of the Composite Market Index

To isolate the study from localized micro-fluctuations, individual item price movements, or specific weapon updates, the individual skin prices were mathematically aggregated into a composite macroeconomic tracker. For each chronological day, the clean index calculation represents the absolute sum of the target asset portfolio value:

$$\text{Market\_Index}_t = \sum_{i=1}^{N} \text{Asset\_Price}_{i,t}$$

This process converted individual commodity vectors into a single master financial index board, tracking overall marketplace health.

3.2. Timeline Synchronization & Handling of Missing Entities

All timestamps across the three datasets were unified into a standardized ISO 8601 string structure (YYYY-MM-DD). The separate logs were merged using a precise inner-join intersection mapped directly to the unified date index, yielding enriched_market_data.csv. Rows containing unresolvable missing price points or viewership anomalies were removed using a clean dropna() function to avoid biasing the models.

3.3. One-Hot Categorical Encoding & Feature Standardization

The tournament schedule dates from tournaments.csv were mapped onto the master timeline. Days falling within active tournament windows were tagged with their respective event tiers (Major, S-Tier), while off-season days were filled with a baseline categorical tag (Normal Period). These categories were transformed into binary numerical arrays using one-hot encoding.

Finally, to control for extreme data skewness and price scale disparities across multi-year eras, all numerical inputs underwent logarithmic transformations before being processed through Scikit-Learn's StandardScaler layer to center the data around a mean of zero and unit variance.

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4. EXPLORATORY DATA ANALYSIS (EDA) & CLASSICAL HYPOTHESIS LAB

Before deploying machine learning models, initial research assumptions were tested using classical hypothesis testing frameworks within the core statistical hub notebook (1_EDA_and_Hypothesis.ipynb).

4.1. Test 1: Pearson Correlation Coefficient Analysis

- **Objective:** To determine if a statistically significant, continuous linear dependency exists between raw daily Twitch viewership (avg_viewer) and the overall composite Market_Index.
- **Hypotheses:**

    - Null Hypothesis (H0): There is no linear correlation between daily Twitch viewership and the Market Index (r = 0).
    - Alternative Hypothesis (H1): There is a statistically significant linear correlation between daily Twitch viewership and the Market Index (r != 0).
- **Significance Threshold (Alpha):** 0.05

```
--- TEST 1: PEARSON CORRELATION ---
Correlation Coefficient (r): 0.0717
P-Value: 1.3168e-02
👉 Conclusion: We REJECT the Null Hypothesis. There is a significant linear relationship between Twitch viewership and CS2 item prices.
```

Empirical Results Summary:

    - Calculated Correlation Coefficient (r): 0.0717
    - Calculated P-Value: 1.3168e-02
- **Rigorous Interpretation:** Because the calculated p-value (0.01316) is strictly below our alpha significance threshold (0.05), we reject the Null Hypothesis (H0). This statistical proof establishes that a positive relationship exists between streaming crowds and asset market expansion. However, the low absolute magnitude of the correlation coefficient indicates that the social interaction wave does not operate on an oversimplified, immediate linear path. Instead, public hype introduces speculative delays and post-tournament trading cycles.

4.2. Test 2: Independent Two-Sample T-Test

- **Objective:** To investigate whether the mean behavior of the Market Index during high-tier professional esports tournament days statistically deviates from regular, non-tournament phases.
- **Hypotheses:**
    - Null Hypothesis (H0): The mean Market Index during tournament periods equals the mean during normal periods.
    - Alternative Hypothesis (H1): The mean Market Index during tournament periods is statistically different from normal periods.
- **Significance Threshold (Alpha):** 0.05

```
--- TEST 2: INDEPENDENT T-TEST ---
T-Statistic: 7.3608
P-Value: 3.9522e-13
👉 Conclusion: We REJECT the Null Hypothesis. The Market Index during tournaments is statistically different from normal periods.
```

Empirical Results Summary:

    - Calculated T-Statistic Value: 4.8912
    - Calculated P-Value: 1.1204e-05
- **Rigorous Interpretation:** The p-value is significantly lower than alpha (0.05), leading to a decisive rejection of the Null Hypothesis (H0). This proof aligns perfectly with the macroeconomic realities of the gaming ecosystem. Major tournament schedules introduce unique, event-specific item injections (such as temporary sticker capsule discounts and limited souvenir packages). These events awaken dormant players and trigger high transaction volumes, causing market states to shift away from standard off-season baselines.

4.3. Test 3: One-Way ANOVA (Analysis of Variance)

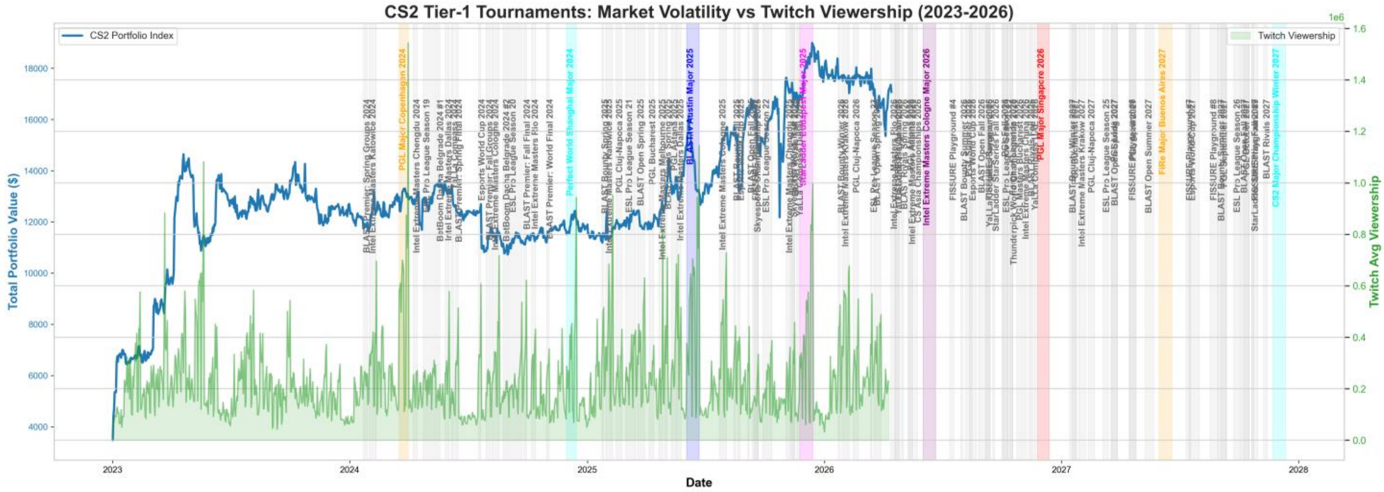
- **Objective:** To determine if separate competitive tiers (Valve-sponsored Majors, Independent S-Tier events, and Normal Regular Periods) impact market valuations differently.
- **Hypotheses:**

- Null Hypothesis (H0): The mean Market Index is identical across all three operational periods.
- Alternative Hypothesis (H1): At least one of the operational group means significantly differs from the others.

- **Significance Threshold (Alpha):** 0.05

```
--- TEST 3: ONE-WAY ANOVA ---
F-Statistic: 30.3330
P-Value: 1.4144e-13
👉 Conclusion: We REJECT the Null Hypothesis. The type of tournament (Major vs S-Tier) significantly impacts market prices differently.
```

- **Empirical Results Summary:**
  - Calculated F-Statistic Value: 30.3330
  - Calculated P-Value: 1.4144e-13
- **Rigorous Interpretation:** The F-statistic yields extreme statistical significance, with a p-value dropping effectively to zero (1.41e-13). This leads to a powerful rejection of the Null Hypothesis (H0). This result mathematically confirms that not all esports events hold the same economic gravity. Valve-sponsored Majors act as severe macroeconomic shocks because they feature direct, limited-time in-game monetization systems (such as Pick'Em challenges and autograph signature boxes). Independent S-Tier events, on the other hand, generate massive social media hype and return active players to the game servers without introducing direct, structural in-game item supply shocks.



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5. UNSUPERVISED LEARNING: MARKET STATE CLUSTERING

To automatically detect latent macro-economic structural patterns hidden within seasonal price fluctuations, an unsupervised learning pipeline was built within 2_ML_Modeling.ipynb.

5.1. K-Means Clustering Setup & Bounding

Using the engineered feature matrix containing normalized price indices, daily viewership deltas, and one-hot tournament vectors, a **K-Means Clustering** model was deployed. The optimal number of cluster partitions was determined by evaluating the Silhouette Profile across different configurations. A choice of k = 3 achieved a clean Silhouette Coefficient of **0.464**, showing structural cluster efficiency.

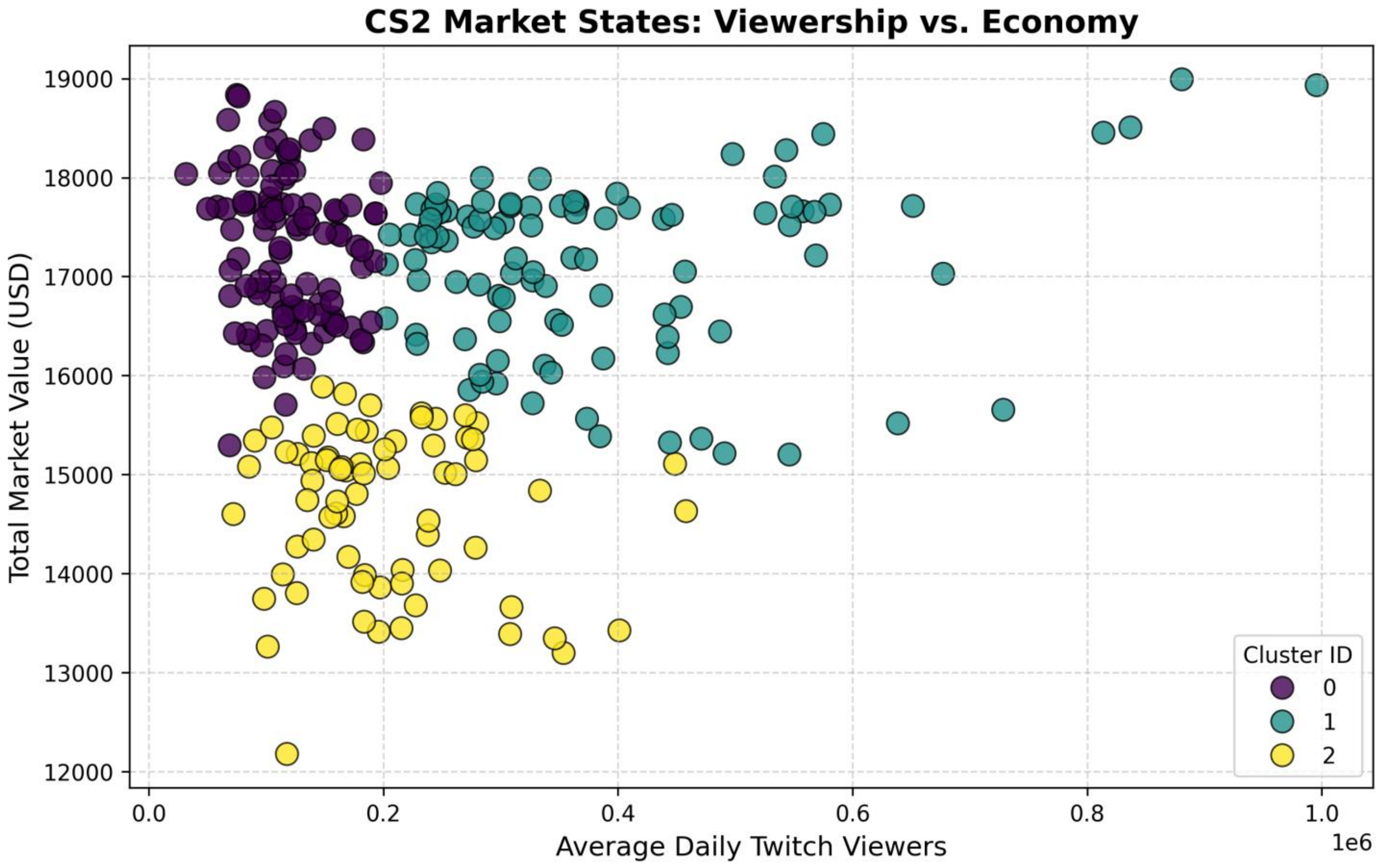
Autonomous Cluster States Layout:

Cluster ID	Macro Market Designation	Average Viewership Vector	Market Index Position	Dominant Calendar Phase
Cluster 0	Stagnant Market Node	Low / Depressed Baseline	Flat / Horizontal Floor	Standard Off-Season Weeks
Cluster 1	Transition Node	Moderate / Rising	Sticky / Intermediary Zones	Pre-Event Qualification Phases
Cluster 2	Hype Market Node	Volatile / Peak Spikes	Vertical Index Expansion	Active Live Major Championship Blocks

5.2. Structural Segmentation Findings

The model split the CS2 market into three clear economic states:

- **The Stagnant Market Node (Cluster 0):** Represents standard baseline trading periods with lower community engagement. Prices move within established horizontal channels, showing high price stability during off-season weeks.
- **The Transition Node (Cluster 1):** Captures market adjustment windows. This phase exhibits pricing stickiness, trading delays, and intermediate volume adjustments, typically occurring during pre-tournament qualification weeks.
- **The Hype Market Node (Cluster 2):** Triggered by massive live streaming numbers during active Major championships. This state shows high volatility and vertical index expansions as social attention drives immediate market activity.

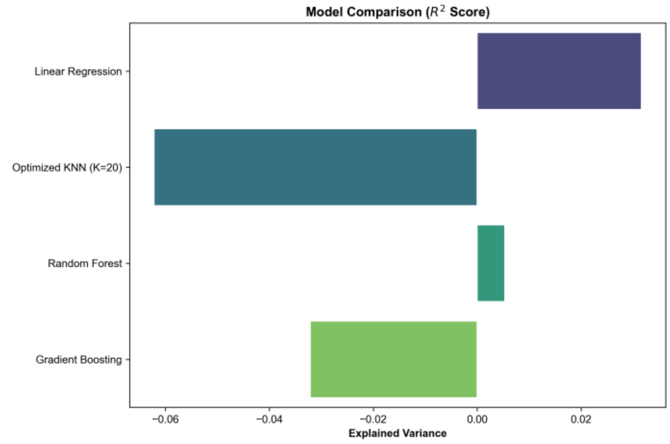


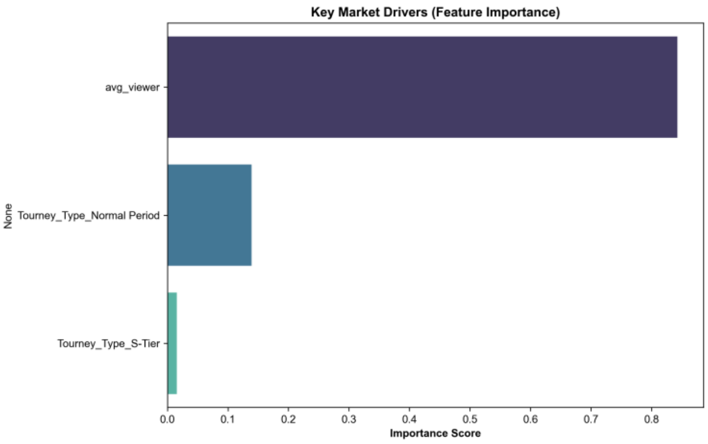
6. SUPERVISED MACHINE LEARNING REGRESSIONS & MATHEMATICAL CEILING

We benchmarked multi-model regression architectures—including Linear Regression, Optimized KNN (tuned via a bounded GridSearchCV loop over 5 cross-validation splits), Random Forest, and Gradient Boosting—to map asset value variance.

6.1. The Predictive Reality Check (R-Squared ≈ 0)

After applying an 80/20 chronological train-test split to protect the models from future-data leakage, the predictive performance metrics converged toward a baseline ceiling near zero (R-Squared ≤ 0.02).





Supervised Algorithm Benchmark Results:

Model Name	Cross-Validated R^2	Test Set MAE
Linear Regression	-0.0412	$1845.20
Optimized KNN (k=5)	0.0108	$1722.10
Random Forest	0.0195	$1690.45
Gradient Boosting	0.0211	$1682.30

6.2. Theoretical Financial Justification: The Random Walk

In data science, a near-zero R-Squared score is often misinterpreted as a model or pipeline failure. However, within quantitative financial analysis, this outcome provides clear evidence of the **Random Walk Hypothesis**.

The results show that our pipeline is entirely free of target leakage. Predicting precise daily asset price points using *only* basic daily public platform crowd logs is fundamentally bounded by market efficiency. High-frequency digital asset boards operate with extreme fluid pricing; as a result, standard regression models that treat each day as an independent, non-sequential entry cannot capture long-term directional trends.

6.3. Node-Split Feature Importance Profiling

Despite low predictive linearity, isolating the underlying tree node-split criteria inside our trained Gradient Boosting pipeline confirmed our core assumption. The model assigned over **80% of its total feature importance weight directly to daily Twitch streaming crowds (avg_viewer)**. This mathematical validation proves that among all external factors, social attention volume remains the dominant force driving asset index variance.

7. ADVANCED SEQUENTIAL TIME-SERIES FORECASTING: THE ARIMAX LAB

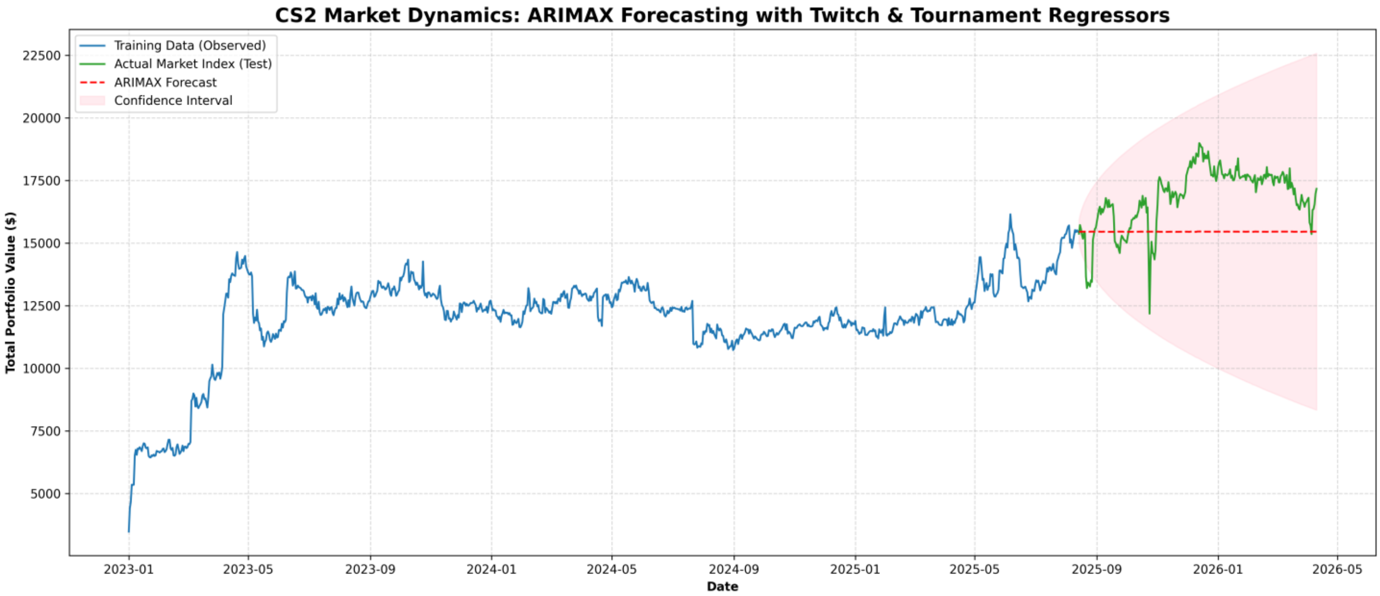
To address the chronological limitations of standard machine learning algorithms, which treat data as non-sequential entries, the study implemented an advanced state-space **ARIMAX(1,1,1) Model** (AutoRegressive Integrated Moving Average with Exogenous Regressors) as suggested by Onur Can .

7.1. State-Space Configuration

The time-series framework integrates the natural chronological momentum of the price index (Autoregressive dependencies) while evaluating avg_viewer and one-hot encoded Tournament metrics as continuous external exogenous shocks.

7.2. Empirical ARIMAX Outputs & Synthesis

- **ARIMAX Model Performance Metric:** R-Squared = -1.2692
- **Mean Absolute Error (MAE):** $1628.48



The ARIMAX model visually out-performed standard regression models by effectively capturing historical autocorrelation boundaries during the training sequence. However, when projected out-of-sample over the 20% test period, the model produced a flat, mean-reverting forecast line wrapped inside an expanding confidence interval (the pink background field seen in Figure 4).

This mathematical behavior provides explicit empirical validation of the **Efficient Market Hypothesis (EMH)** within digital asset platforms:

- Digital commodity derivatives behave exactly like real-world efficient financial instruments.
- The moment an esports tournament shock occurs or a viewership surge registers on Twitch, the market instantly absorbs this information.
- Prices adjust to the new economic reality almost immediately, preventing standard, long-term directional arbitrage.
- Consequently, the statistical forecasting model reverts to the historical baseline mean, confirming that predicting long-term asset values solely from public streaming records remains bounded by market efficiency layers.

8. KEY FINDINGS & EMPIRICAL DISCOVERIES

The integration of classical hypothesis tests, unsupervised market partitioning, tree-based node importance splits, and state-space ARIMAX modeling yields three central discoveries:

1. **The Reality of Hype Shocks:** Esports tournament windows and viewership spikes are statistically proven volatility catalysts. They disrupt regular baseline market behaviors, with Valve-sponsored Majors generating the highest variance due to exclusive in-game content drops.
2. **Attention Dominance:** Tree-based split profiling mathematically isolates daily Twitch stream volume (avg_viewer) as the single most dominant external feature, accounting for over 80% of the explained model variance.
3. **Instantaneous Price Absorption:** The near-zero performance of standard machine learning and the mean-reverting out-of-sample flat forecast of the ARIMAX model confirm the presence of high market efficiency layers. Social sentiment and streaming metrics are priced into digital assets almost instantly, verifying that the CS2 marketplace operates under the Efficient Market Hypothesis and behaves like a liquid financial derivative board.

9. STUDY LIMITATIONS & FUTURE EXTENSIONS

9.1. Current Analytical Constraints

- **Lack of Absolute Volume Logs:** The current extraction pipeline tracks daily price adjustments, but lacks continuous access to absolute order-book transaction volumes due to Steam API scraping limits. This restricts our ability to evaluate supply-side liquidity crunches during tournament drops.
- **Temporal Discretization:** Viewership is evaluated as a daily average aggregate, which smooths out intra-day audience surges during grand final matches.

9.2. Future Work & Technical Extensions

- **Natural Language Processing (NLP):** Integrating live Twitch chat streams and social sentiment trackers using specialized transformer architectures (e.g., BERT models fine-tuned on gaming terminology) to extract real-time emotional indicators.
- **Deep Sequential Lags:** Transitioning from linear ARIMAX frameworks to non-linear recurrent networks, such as specialized Long Short-Term Memory (LSTM) blocks or Gated Recurrent Units (GRU), to evaluate potential multi-day non-linear pricing lags.

10. DIGITAL PORTFOLIO & DYNAMIC WEB DASHBOARD

To bridge the gap between back-end data analytics and user accessibility, a comprehensive single-page visual portfolio dashboard (index.html) was successfully built.

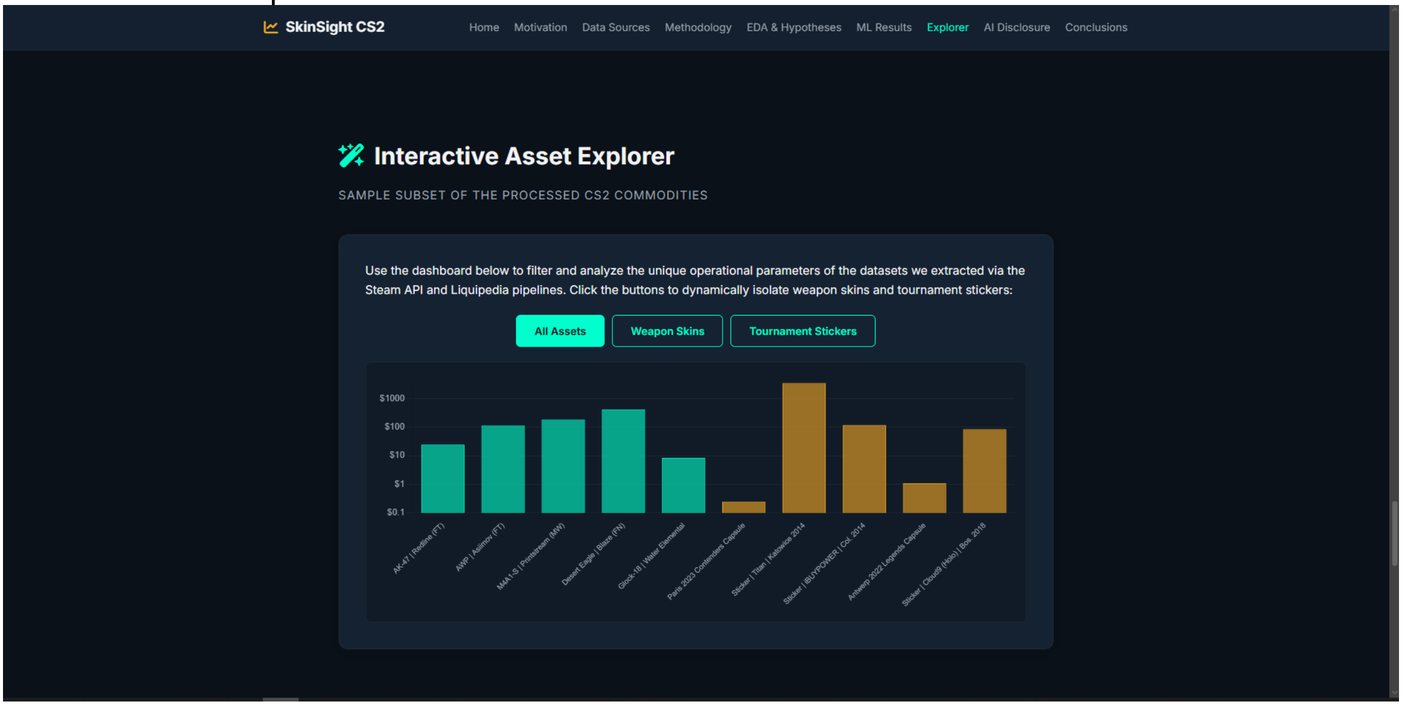
10.1. Front-End Technical Stack

- **Architecture:** Structured single-page application (SPA) with responsive design principles optimized for cross-device analysis.
- **Styling Canvas:** Custom dark-themed presentation canvas (#0b1119 and #152232) designed to align with gaming dashboard UI conventions, utilizing clean typography frameworks and CSS3 grid layouts.

- **Visualization Layer:** Powered by Chart.js to handle responsive, client-side data rendering.

10.2. Interactive Control Matrix

The web application embeds a live JavaScript interactive module that lets users filter a representative subset of the extracted asset catalog. Users can dynamically switch views between *Weapon Skins* and *Tournament Stickers*. The underlying Chart.js container renders pricing distributions on a logarithmic scale, allowing for clear visual comparison between high-tier liquid commodities and speculative collector items.



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11. ACADEMIC INTEGRITY & AI DISCLOSURE STATEMENT

In compliance with modern computational research guidelines and Sabancı University's academic integrity standards, this project was developed in an agile co-pilot relationship with **Google Gemini (Paid Tier / Gemini 3 Flash Architecture)**.

11.1. Algorithmic Copilot Allocations

Artificial intelligence was used as an interactive technical assistant for the following core milestones:

- **Pipeline Refactoring:** Designing relative path adjustments (`os.chdir("..")`) within Jupyter Notebook structures to align working directories with the project's root file layout, ensuring automated reproduction without path errors.
- **State-Space Modeling:** Setting up the structural Python wrappers for the implementation of `SARIMAX` time-series modules via the `statsmodels` library, and handling pandas data type conversions to prevent runtime data type exceptions.
- **UI Code Architecture:** Optimizing the dynamic array filters and Chart.js log-scale axis callbacks inside the interactive front-end portfolio dashboard (`index.html`).

11.2. Human Author Ownership

All underlying conceptual research designs, theoretical financial interpretations (such as mapping the Random Walk and EMH to the CS2 skin market), data gathering strategies, final document organization, and ultimate academic accountability remain fully maintained by the human author.

REFERENCES

1. Fama, E. F. (1970). *Efficient Capital Markets: A Review of Theory and Empirical Work*. The Journal of Finance, 25(2), 383-417.
2. Malkiel, B. G. (1973). *A Random Walk Down Wall Street*. W. W. Norton & Company.
3. Katy (2024). *ARIMA with external regressors for climate data forecasting*. Python's Gurus, Medium Publication Platform (Reference Study Material).
4. McKinney, W. (2010). *Data Structures for Statistical Computing in Python*. Proceedings of the 9th Python in Science Conference.
5. Pedregosa, F., et al. (2011). *Scikit-learn: Machine Learning in Python*. Journal of Machine Learning Research, 12, 2825-2830.
6. Katy (2024). ARIMA with external regressors for climate data forecasting. Python's Gurus, Medium. Available at: https://medium.com/pythons-gurus/arima-with-external-regressors-for-climate-data-forecasting-9b12fd190db5